디지털논리회로설계

Digital Logic Circuit Design

(VHDL Code Result Report)

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major : 로봇공학과 (Robotics)

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Course Number : 23278

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1. VHDL code

-1. RS Latch

```
LIBRARY ieee;
USE ieee.std_logic_1164.ALL;

ENTITY rs_latch IS
 PORT (
 s : IN std_logic; -- Set
 r : IN std_logic; -- Reset
 q : OUT std_logic;
 qb : OUT std_logic
 );
END rs_latch;

ARCHITECTURE bhv OF rs_latch IS
 SIGNAL q_int, qb_int : std_logic;
BEGIN
 q_int <= r NOR qb_int;
 qb_int <= s NOR q_int;

 q <= q_int;
 qb <= qb_int;
END bhv;
```

-2. D Latch

```
LIBRARY ieee;
USE ieee.std_logic_1164.ALL;

ENTITY d_latch IS
PORT (
 clk : IN std_logic;
 d : IN std_logic;
 q : OUT std_logic
);
END d_latch;

ARCHITECTURE bhv OF d_latch IS
BEGIN

 PROCESS(clk, d)
 BEGIN
 IF (clk = '1') THEN
 q <= d;
 END IF;
 END PROCESS;

END bhv;
```

-3. D Flip-Flop

```
LIBRARY ieee;
USE ieee.std_logic_1164.ALL;

ENTITY d_flipflop IS
 PORT (
 clk : IN std_logic;
 d : IN std_logic;
 q : OUT std_logic
 );
END d_flipflop;

ARCHITECTURE bhv OF d_flipflop IS
BEGIN
 process(clk)
 BEGIN
 IF (clk'event AND clk = '1') THEN
 q <= d;
 END IF;
 END process;
END bhv;
```

-4. JK Flip-Flop

```
LIBRARY ieee;
USE ieee.std_logic_1164.all;
USE ieee.numeric_std.all;

ENTITY jk_flipflop IS
 PORT (
 clk : IN std_logic; -- Clock signal
 j : IN std_logic; -- Data input
 k : IN std_logic; -- Data input
 q : OUT std_logic -- Data output
 );
END ENTITY jk_flipflop;

ARCHITECTURE bhv OF jk_flipflop IS
 SIGNAL q_internal : std_logic := '0';
BEGIN
 q <= q_internal;
 process(clk)
 begin

 if (clk'event and clk = '1') then
 if (j = '1' and k = '0') then
 q_internal <= '1'; -- Set output to 1
 elsif (j = '0' and k = '1') then
 q_internal <= '0'; -- Reset output to 0
 elsif (j = '1' and k = '1') then
 q_internal <= not q_internal;
 end if;
 end if;
 end process;
END ARCHITECTURE bhv;
```

-5. D Flip-Flop with Asynchronous Set and Clear

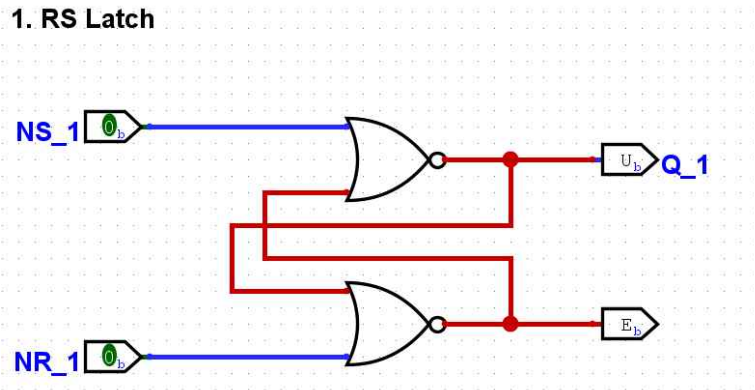
```
LIBRARY ieee;
USE ieee.std_logic_1164.all;
USE ieee.numeric_std.all;
```

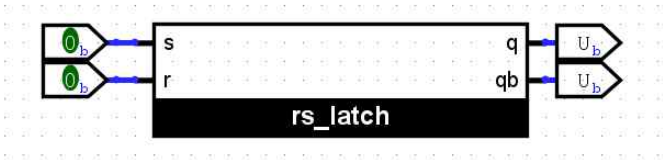
```
ENTITY jkff IS
 PORT (
 clk : IN std_logic; -- Clock signal
 j : IN std_logic; -- Data input
 k : IN std_logic; -- Data input
 clr : IN std_logic; -- lear signal
 set : IN std_logic; -- set signal
 q : OUT std_logic -- Data output
 );
END ENTITY jkff;

ARCHITECTURE bhv OF jkff IS
 SIGNAL q_internal : std_logic := '0';
BEGIN
 -- Drive the output port q with the internal signal
 q <= q_internal;
 process(clk, clr, set)
 begin
 if clr = '1' then
 q_internal <= '0';
 elsif set = '1' then
 q_internal <= '1';
 elsif (clk'event and clk = '1') then
 if (j = '1' and k = '0') then
 q_internal <= '1'; -- Set output to 1
 elsif (j = '0' and k = '1') then
 q_internal <= '0'; -- Reset output to 0
 elsif (j = '1' and k = '1') then
 q_internal <= not q_internal; -- Toggle the output
 end if;
 end if;
 end process;
END ARCHITECTURE bhv;
```

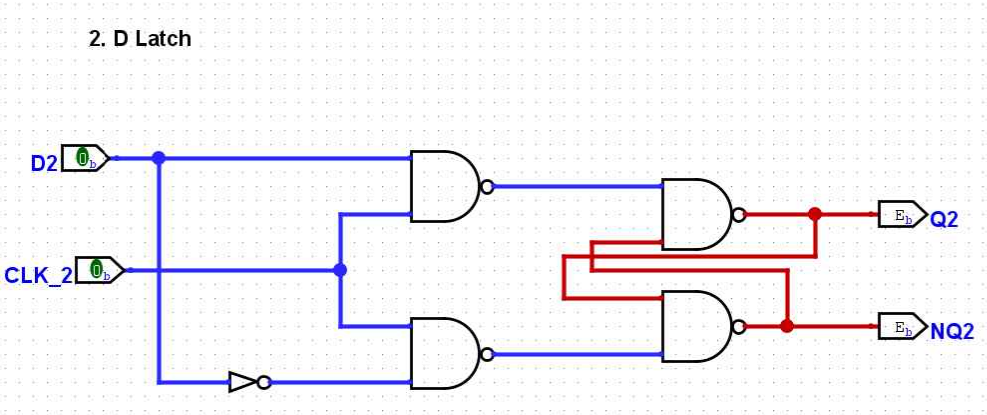
2. Circuit

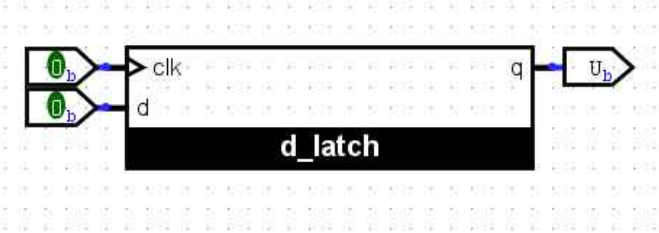
-1. RS Latch



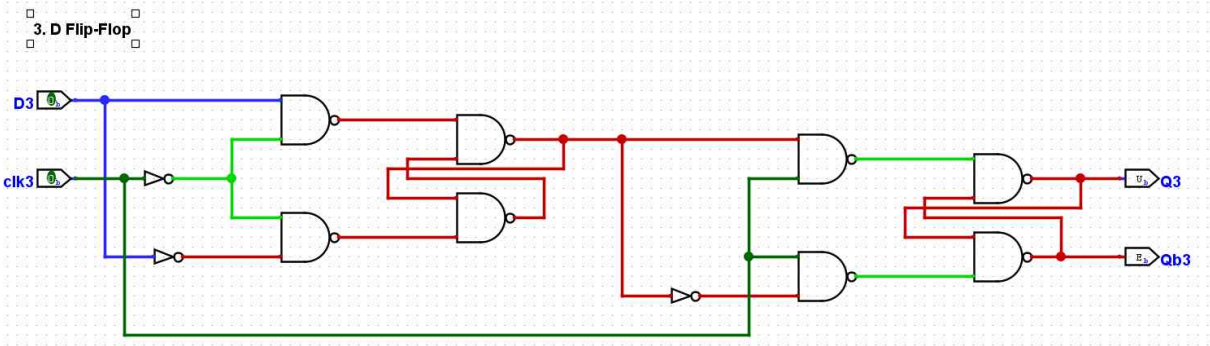


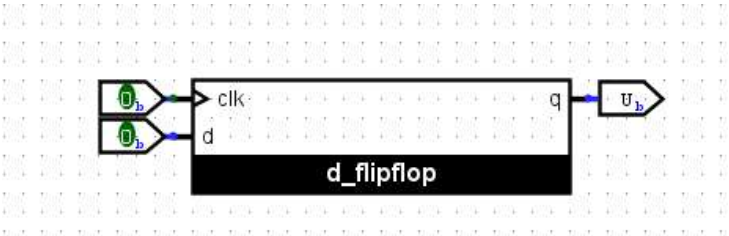
-2. D Latch



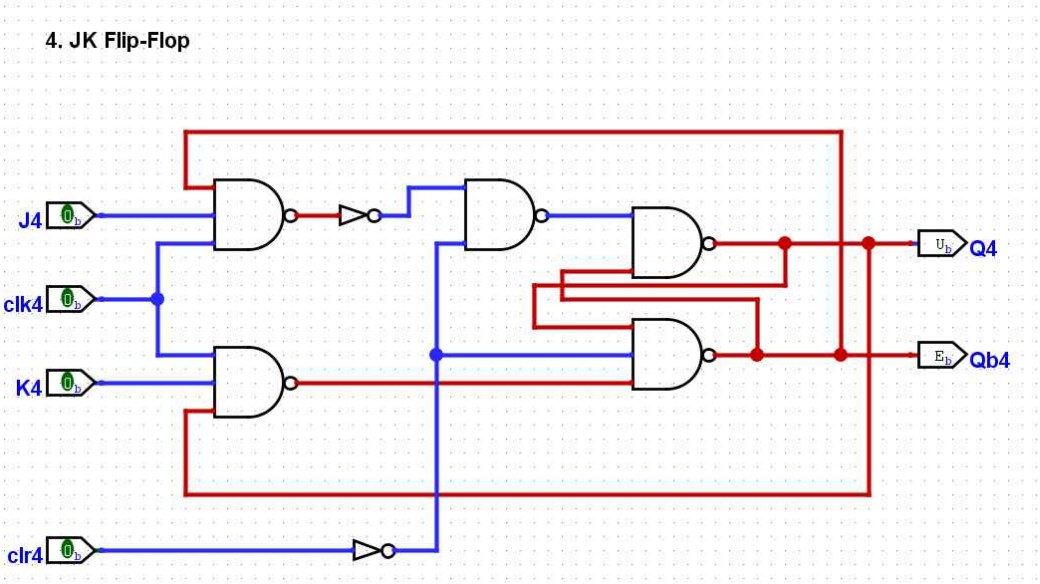


-3. D Flip-Flop



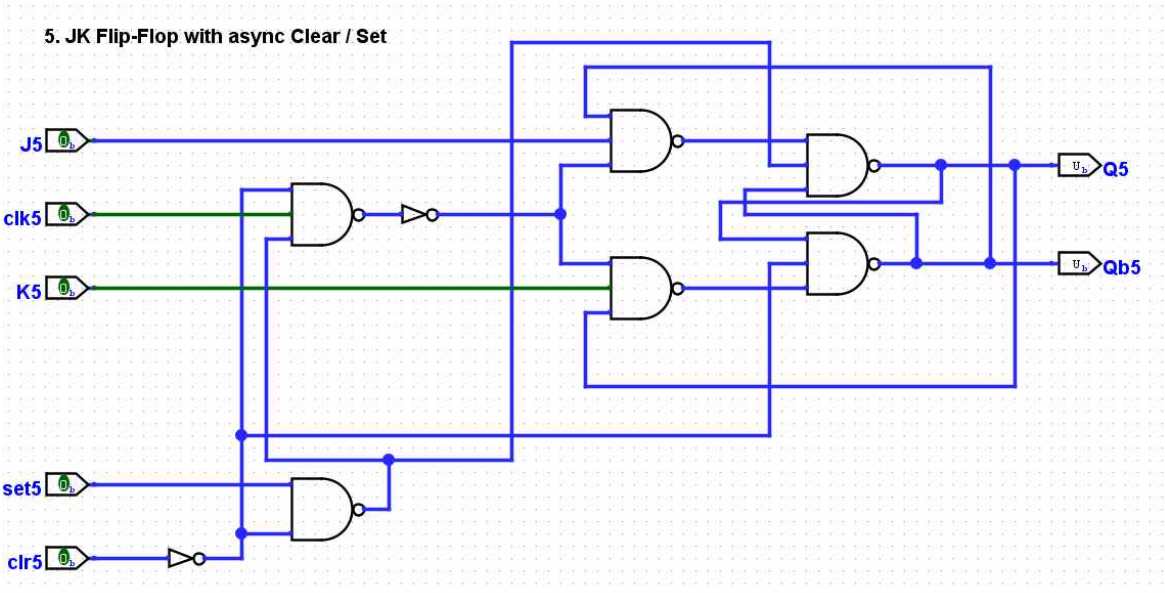


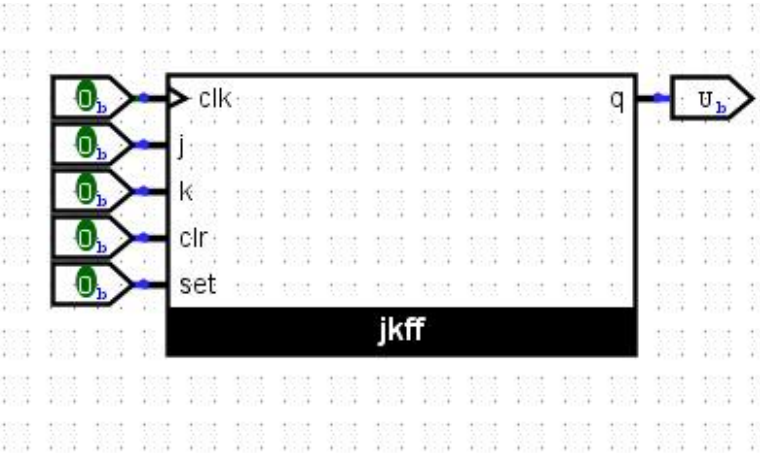
-4. JK Flip-Flop





-5. D Flip-Flop with Asynchronous Set and Clear

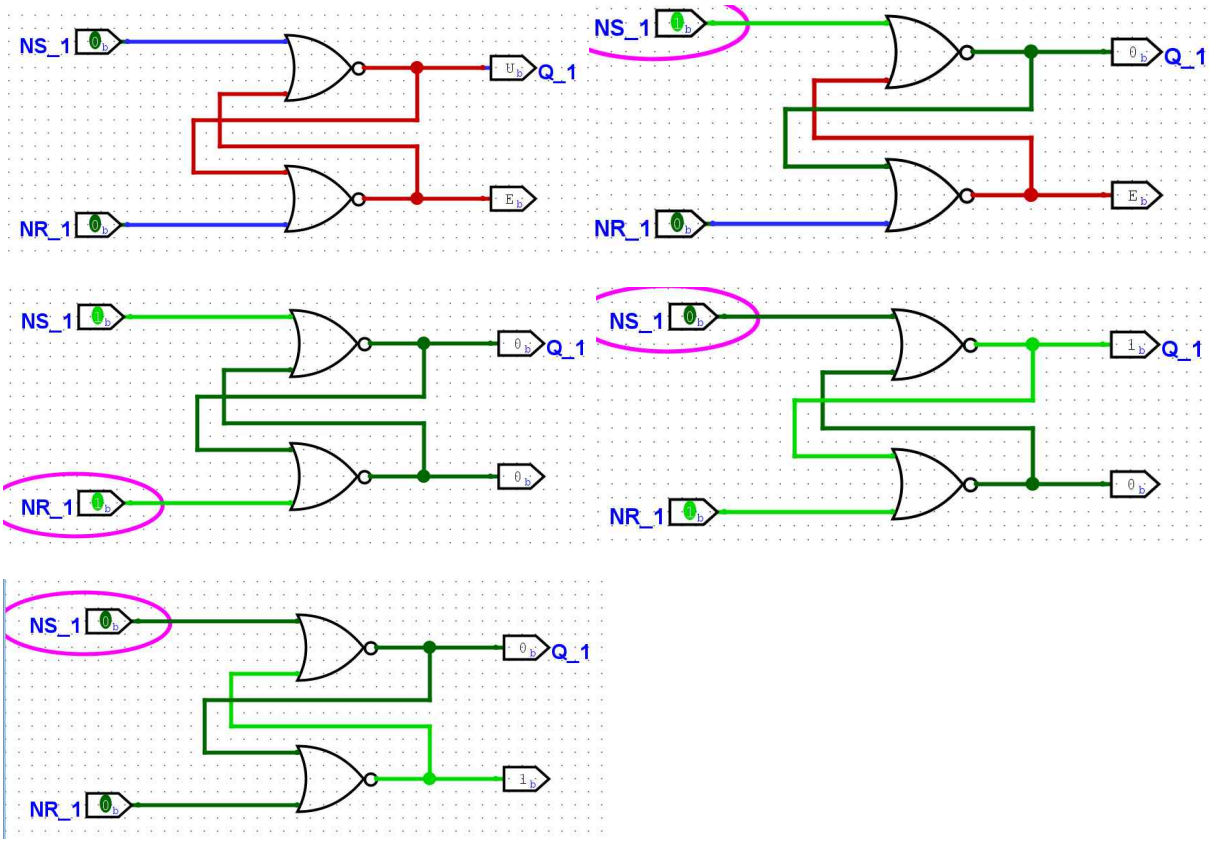




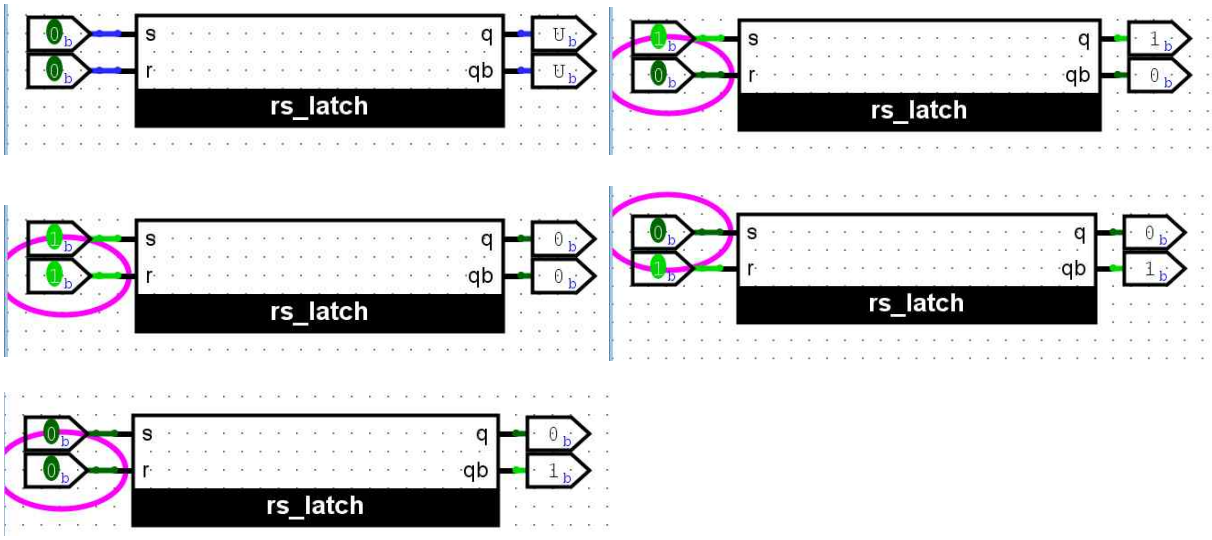
3. Simulation screenshot()

-1. RS Latch

(1) Gate-Level

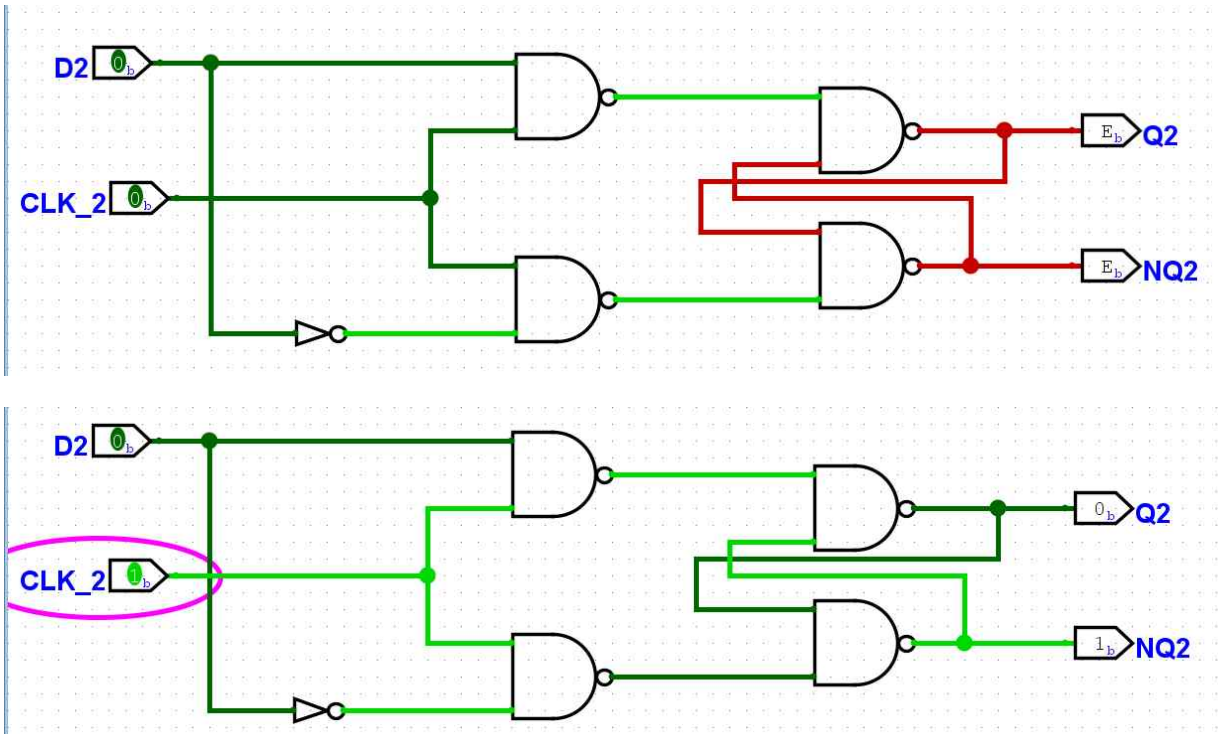


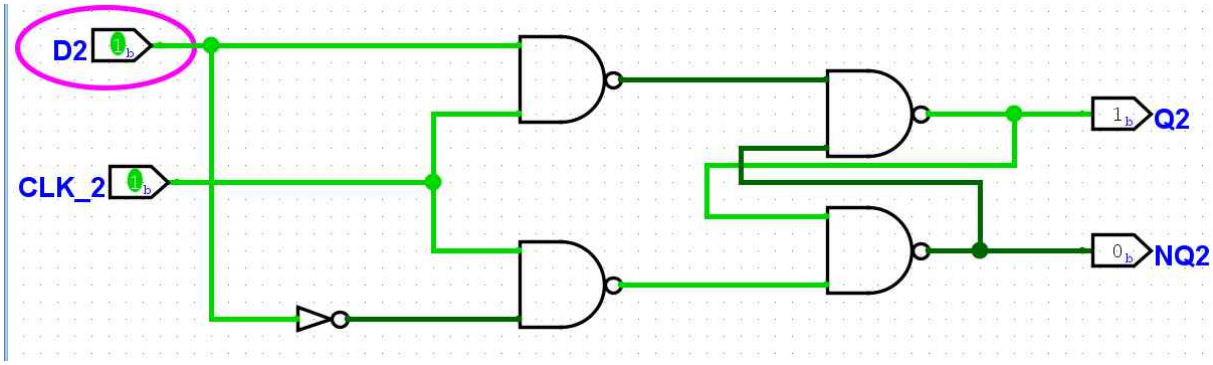
(2) VHDL-Based

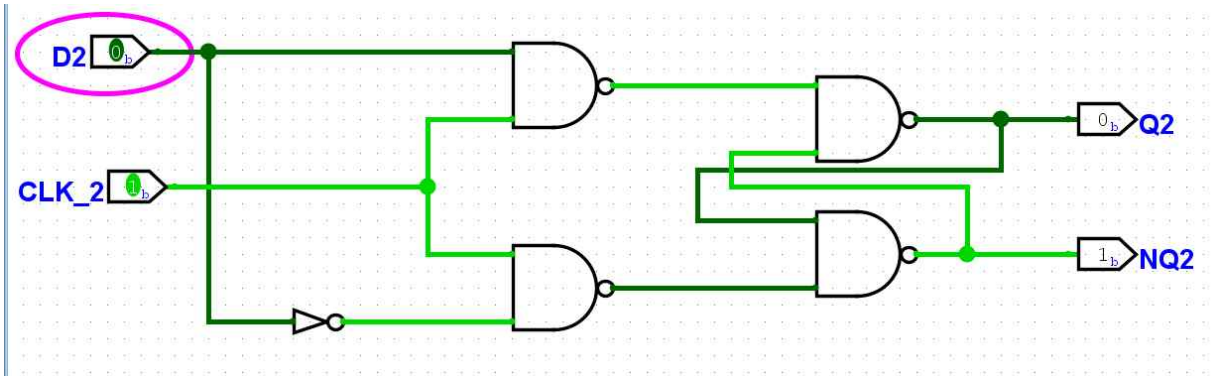


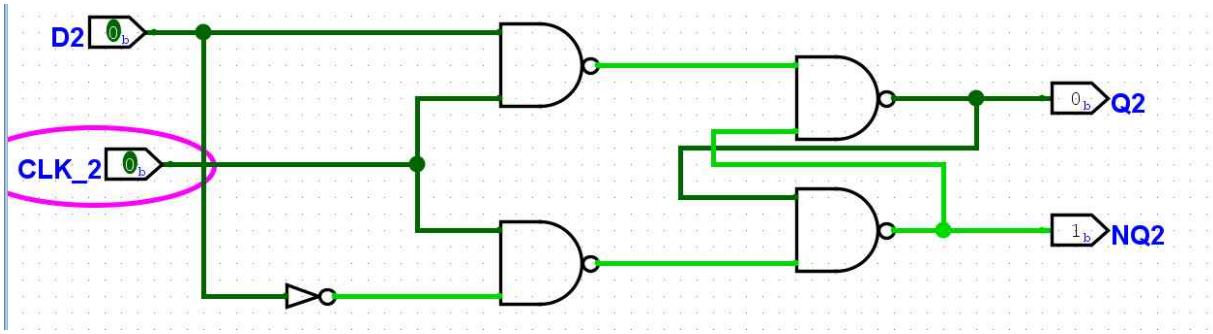
-2. D Latch

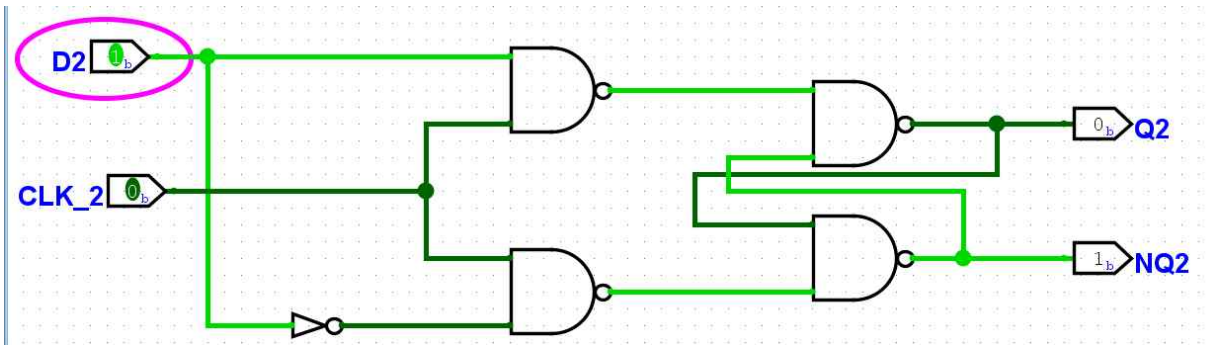
(1) Gate-Level

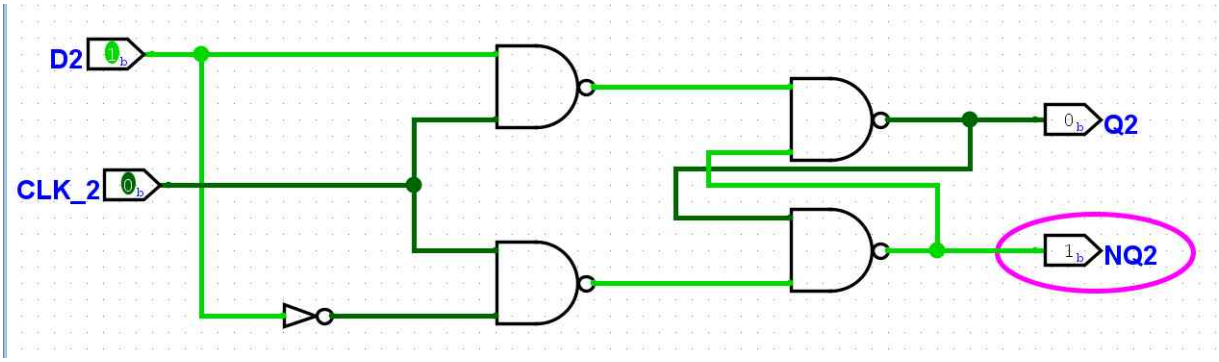


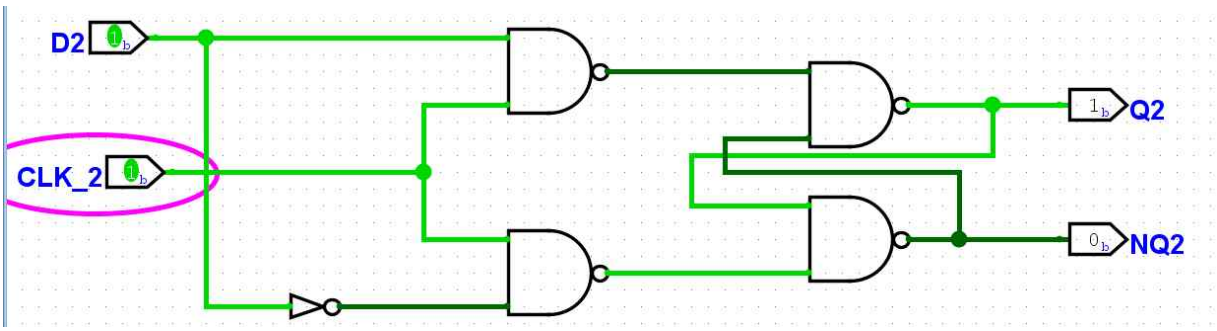

D2
CLK_2
Q2
NQ2

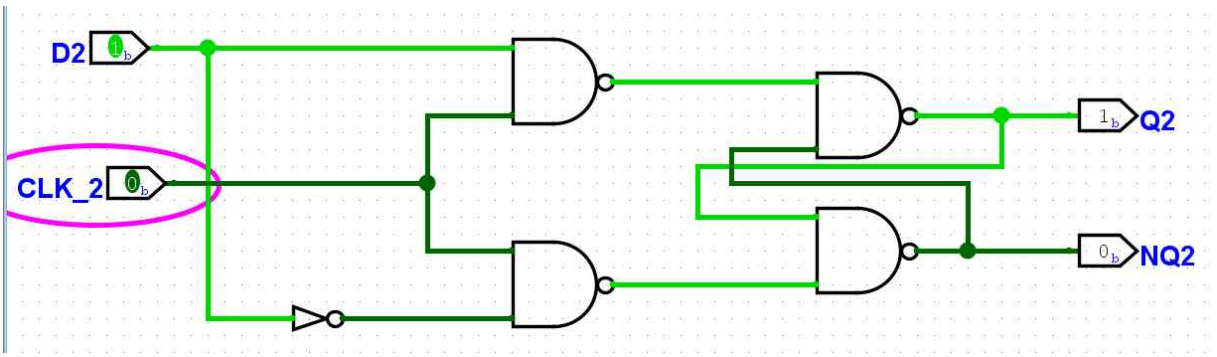

D2
CLK_2
Q2
NQ2

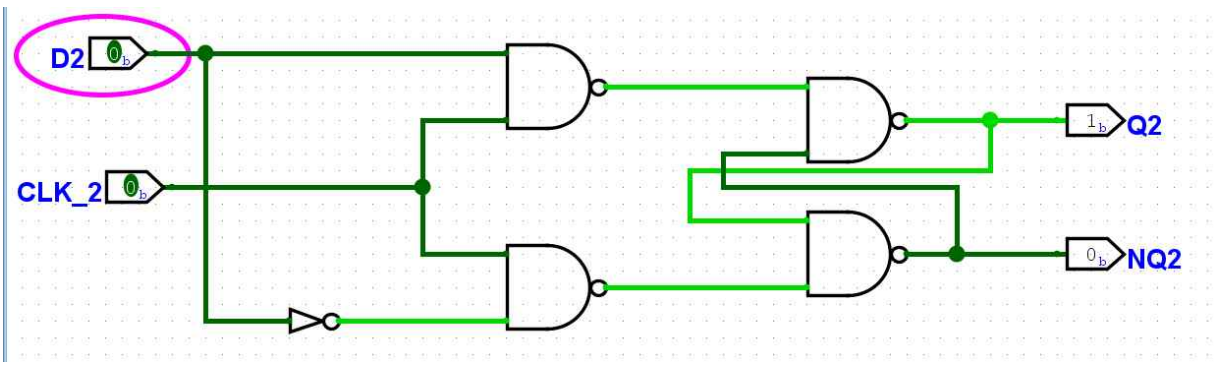

D2
CLK_2
Q2
NQ2


D2
CLK_2
Q2
NQ2

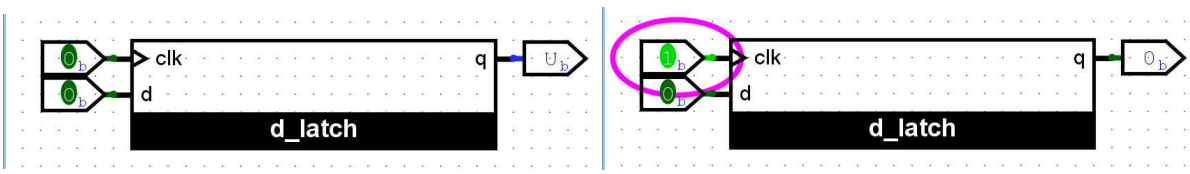

D2
CLK_2
Q2
NQ2

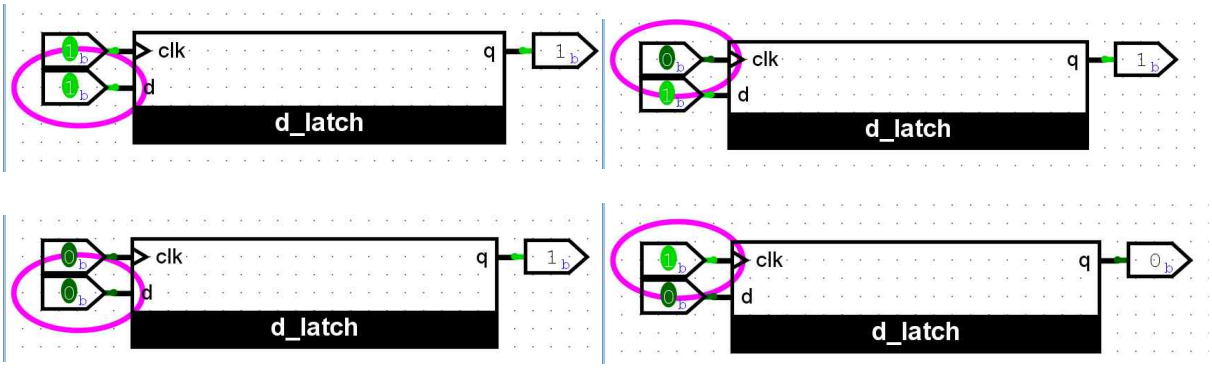

D2
CLK_2
Q2
NQ2


D2
CLK_2
Q2
NQ2


D2
CLK_2
Q2
NQ2

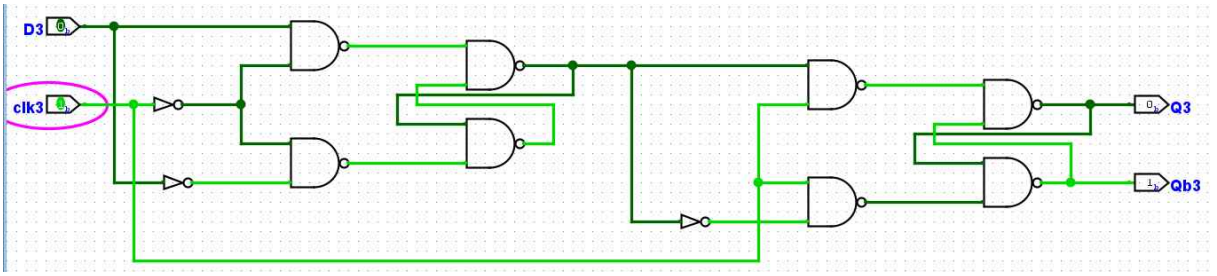
(2) VHDL-Based

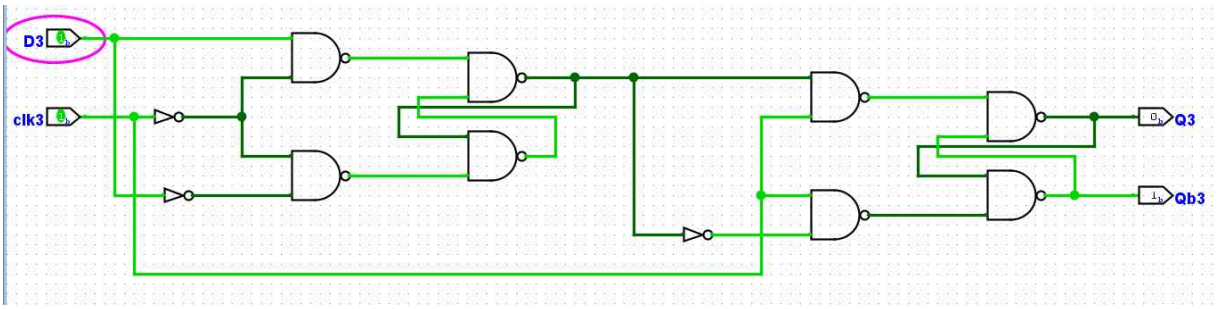

clk
d
q
d_latch
clk
d
q
d_latch

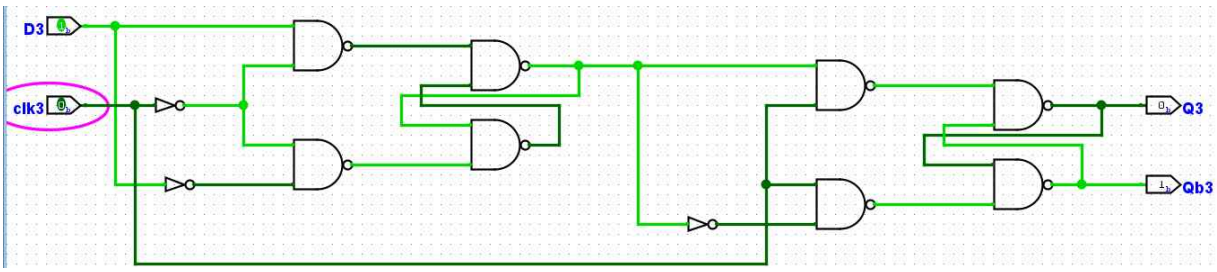


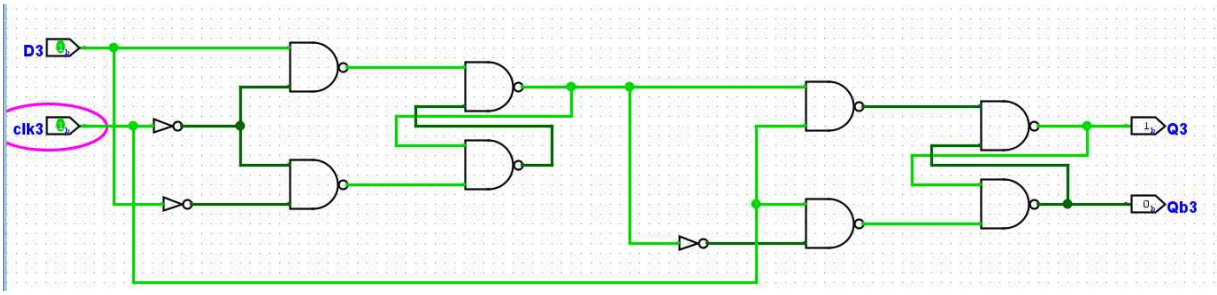
-3. D Flip-Flop

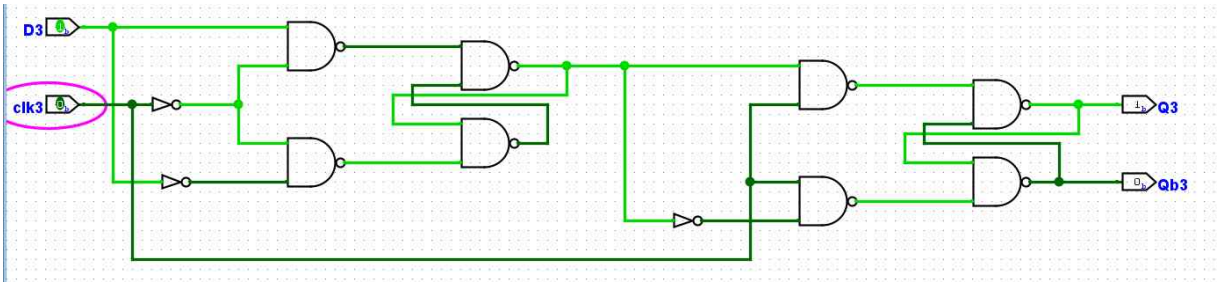
(1) Gate-Level

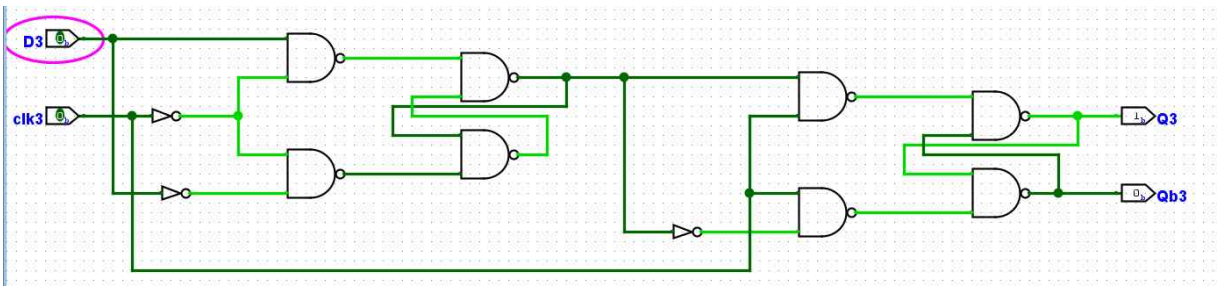


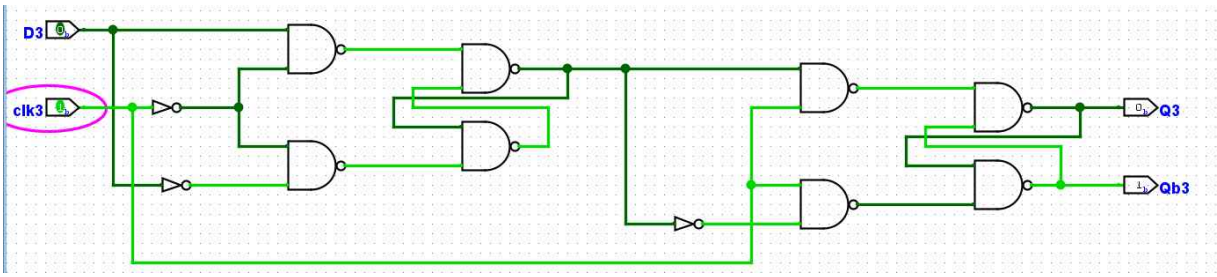




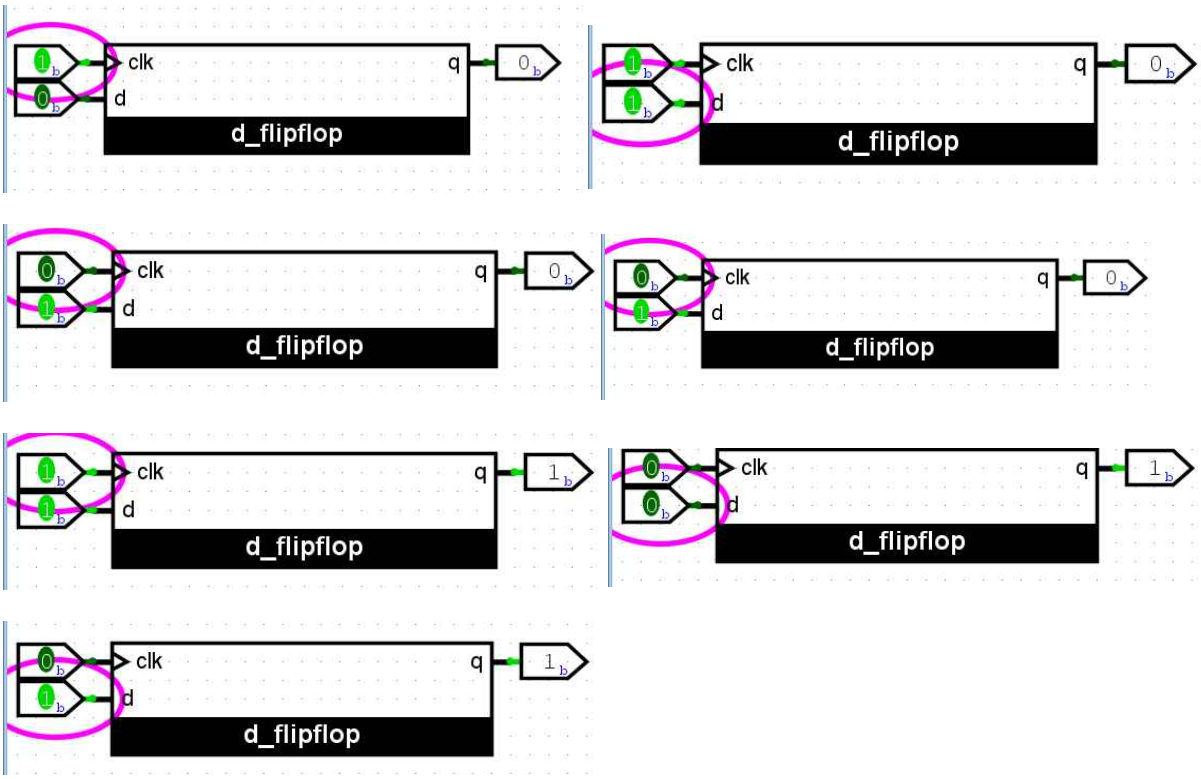






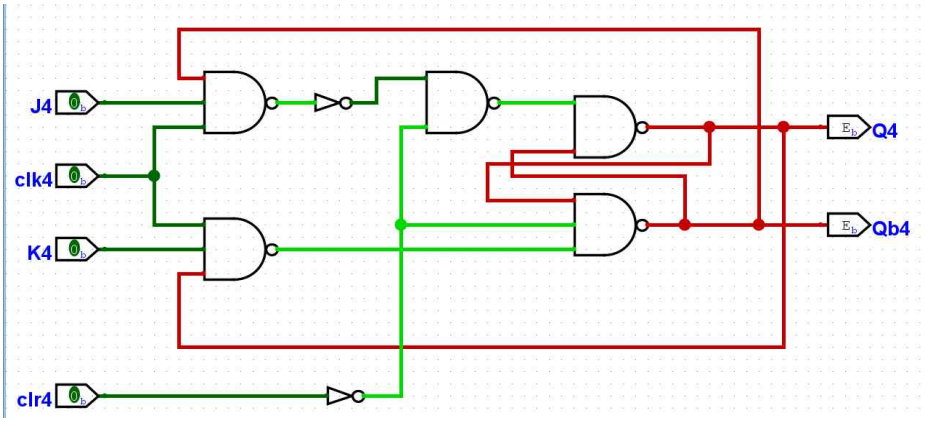


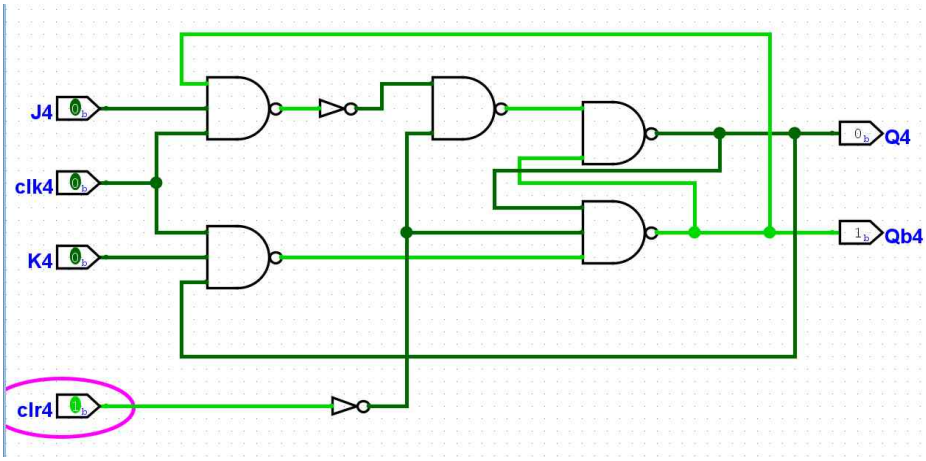
(2) VHDL-Based

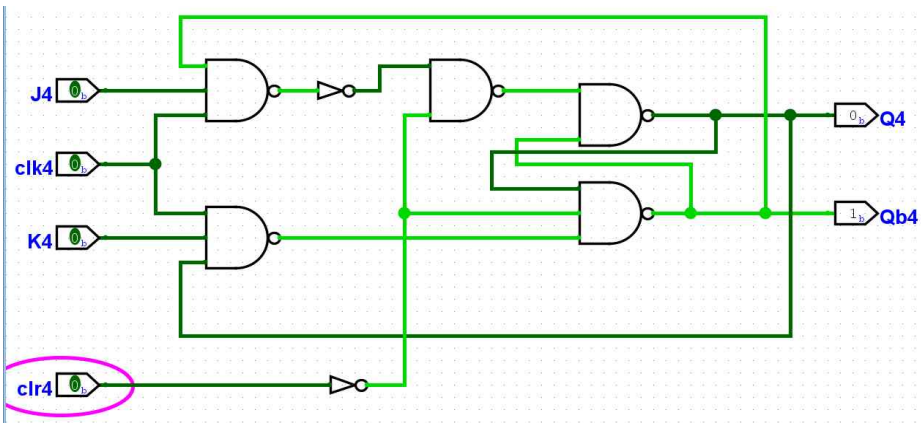


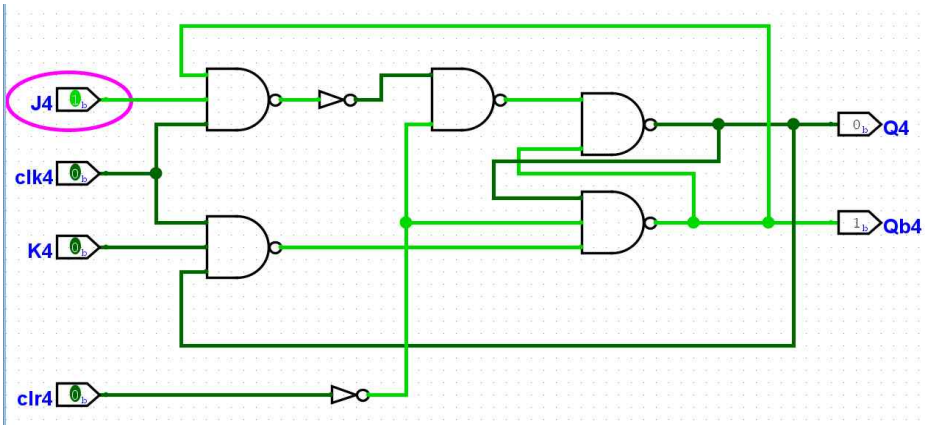
-4. JK Flip-Flop

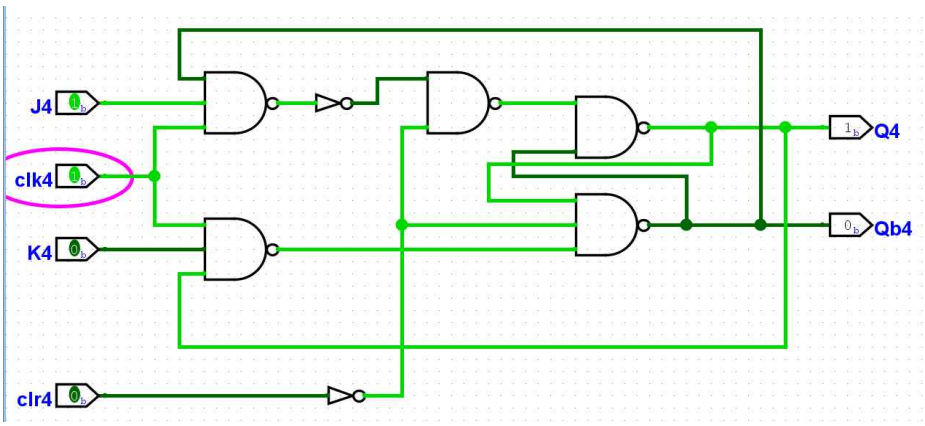
(1) Gate-Level

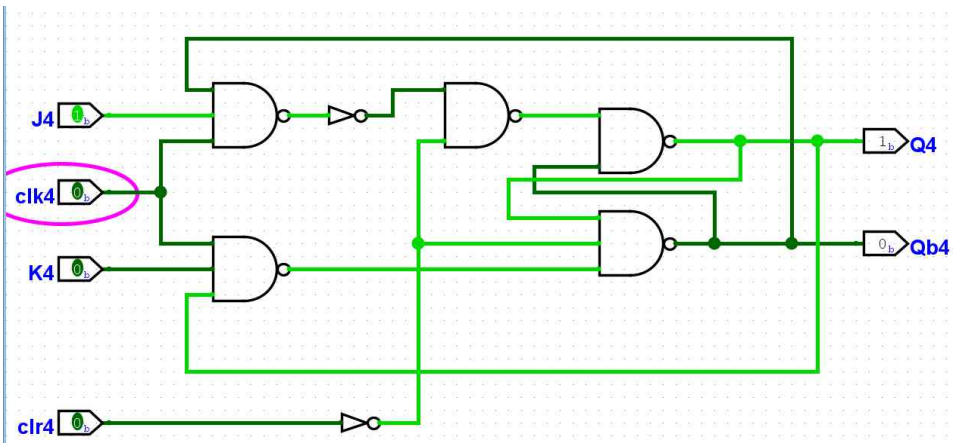


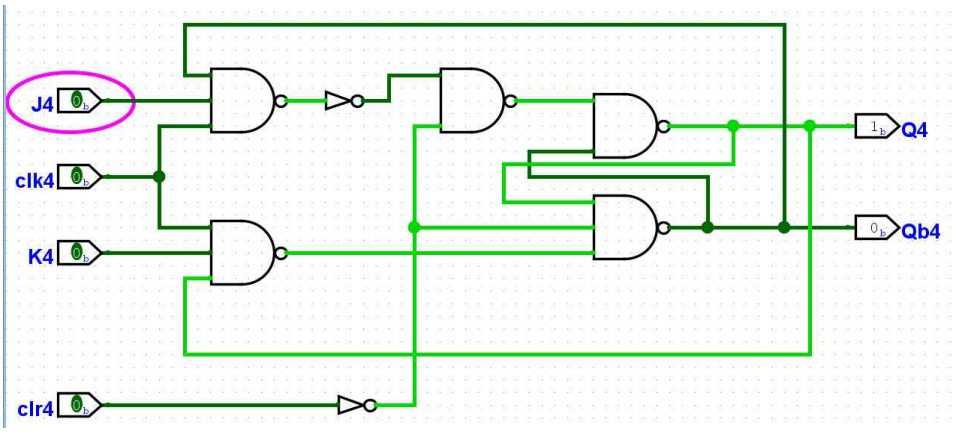


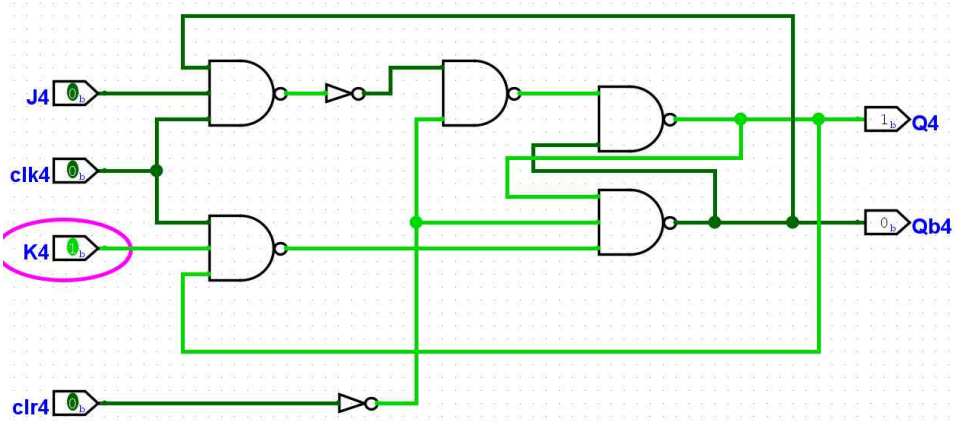


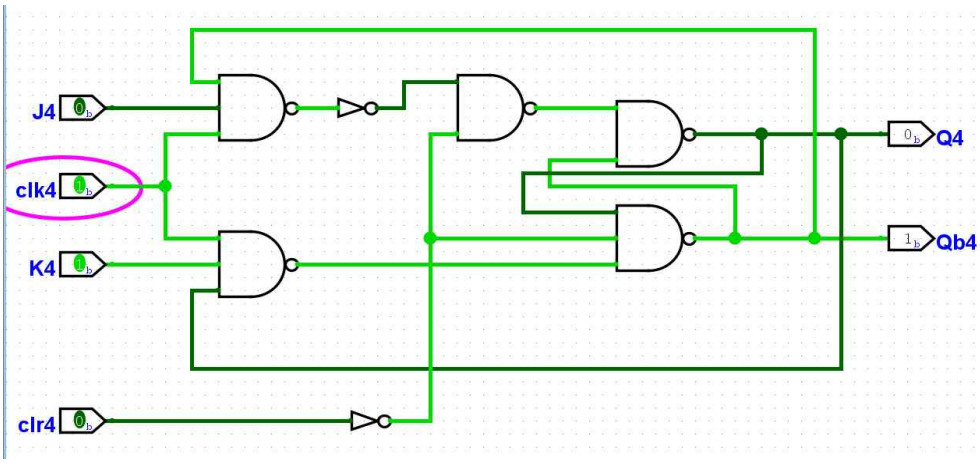

J4
clk4
K4
clr4
Q4
Qb4

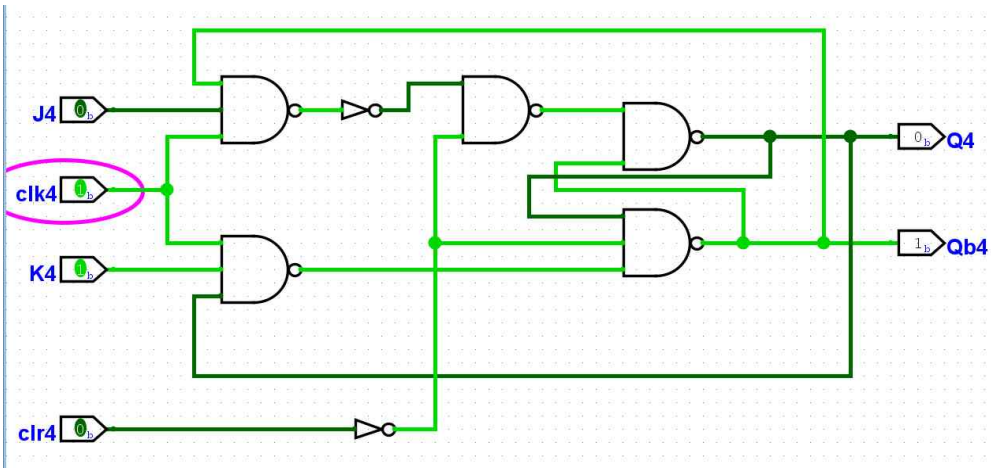

J4
clk4
K4
clr4
Q4
Qb4

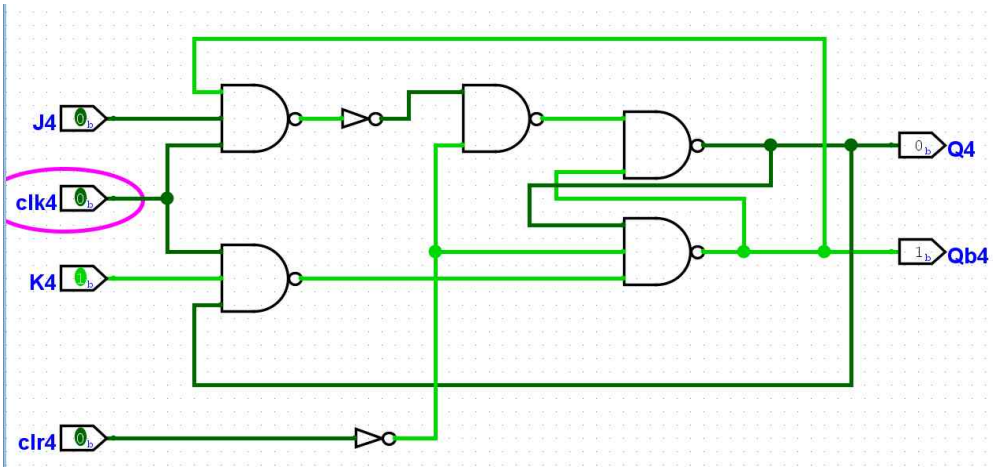

J4
clk4
K4
clr4
Q4
Qb4

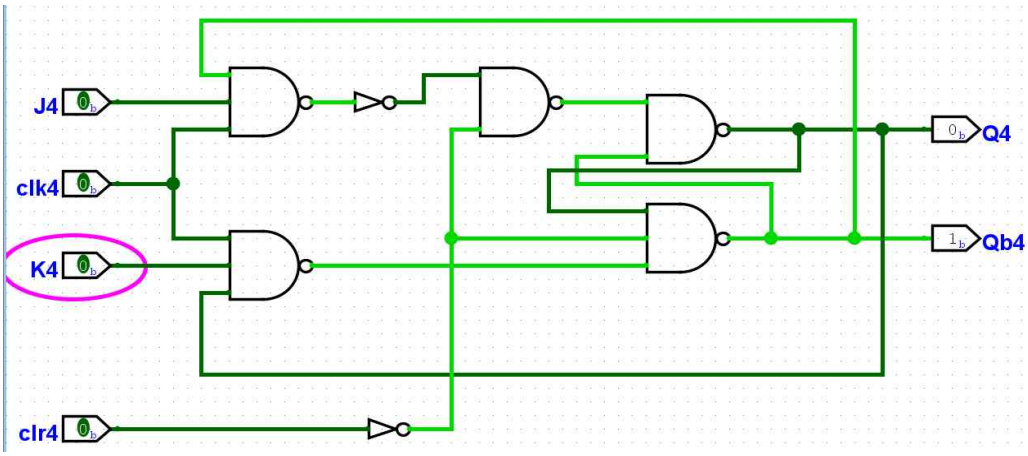

J4
clk4
K4
clr4
Q4
Qb4


J4
clk4
K4
clr4
Q4
Qb4

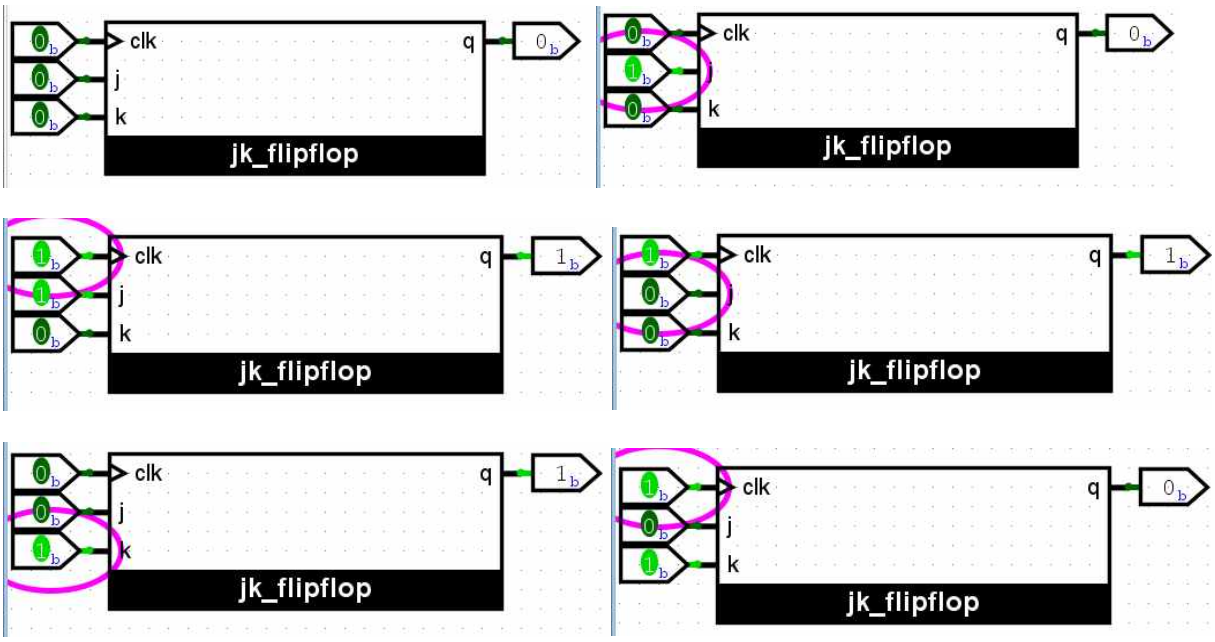

J4
clk4
K4
clr4
Q4
Qb4


J4
clk4
K4
clr4
Q4
Qb4


J4
clk4
K4
clr4
Q4
Qb4


J4
clk4
K4
clr4
Q4
Qb4

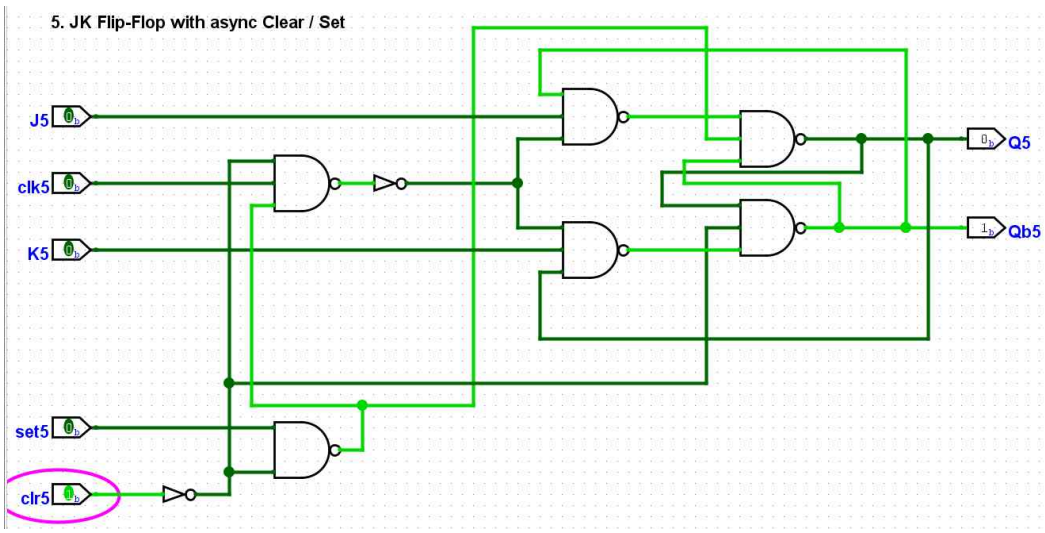
(2) VHDL-Based

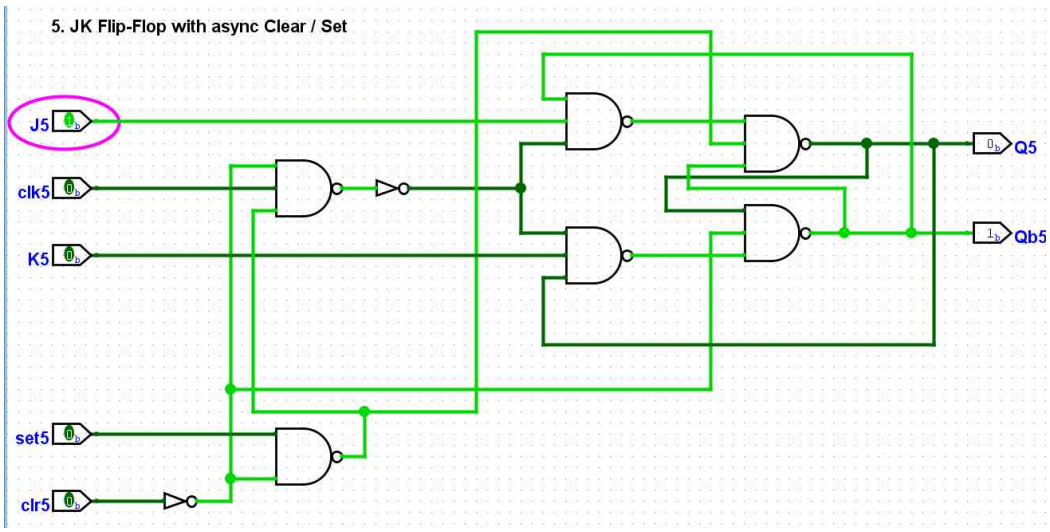

clk
j
k
q
jk_flipflop

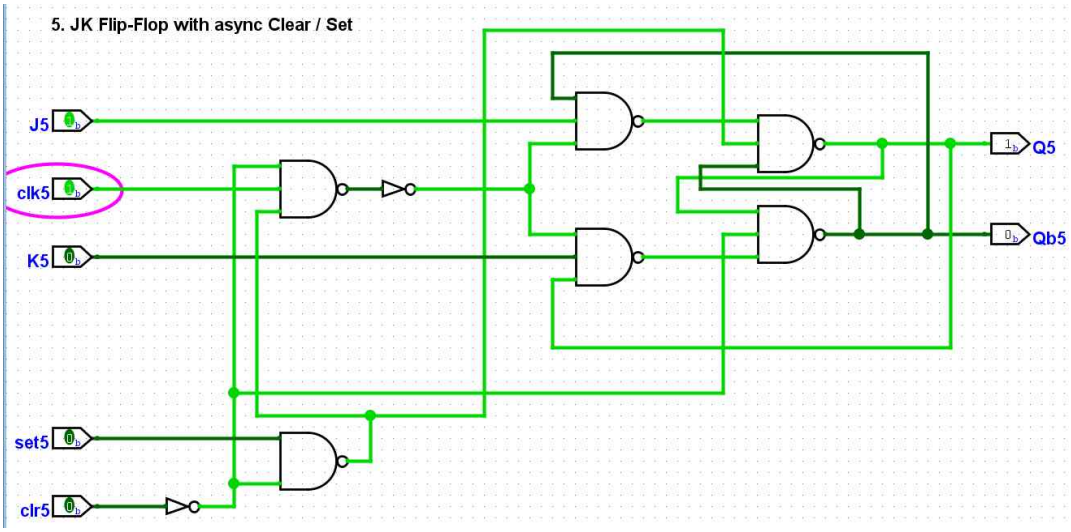


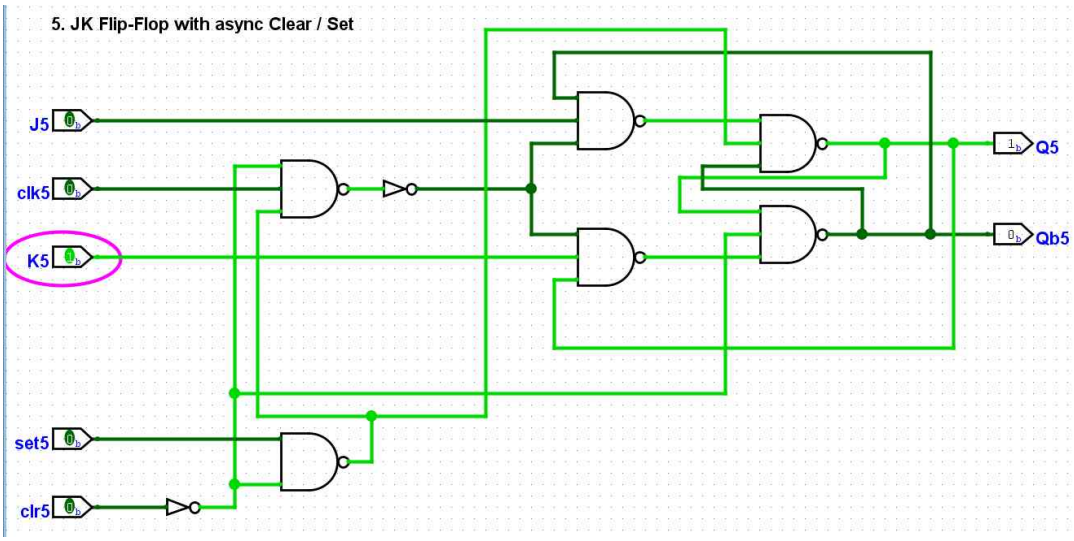
-5. D Flip-Flop with Asynchronous Set and Clear

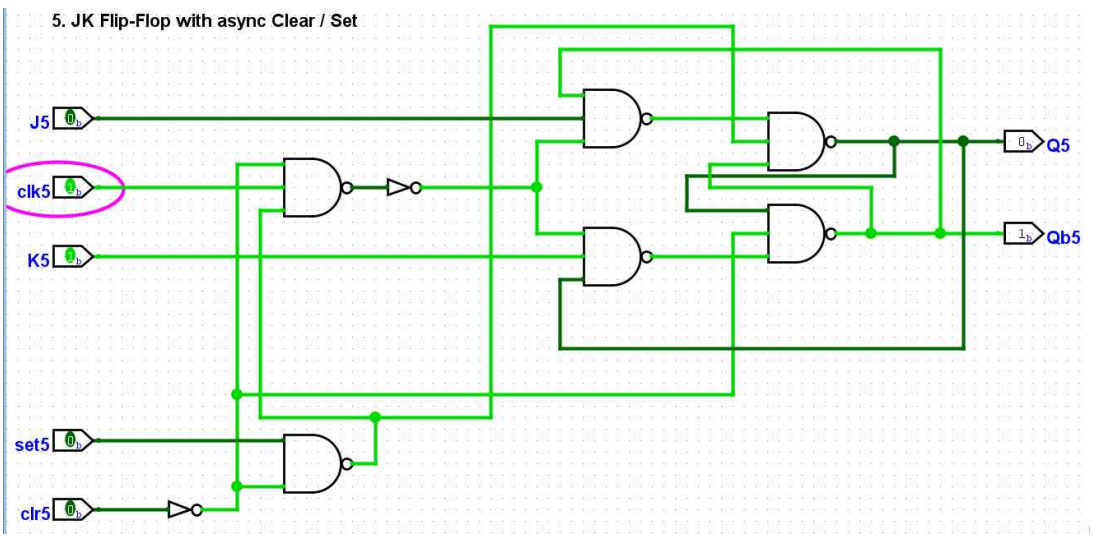
(1) Gate-Level

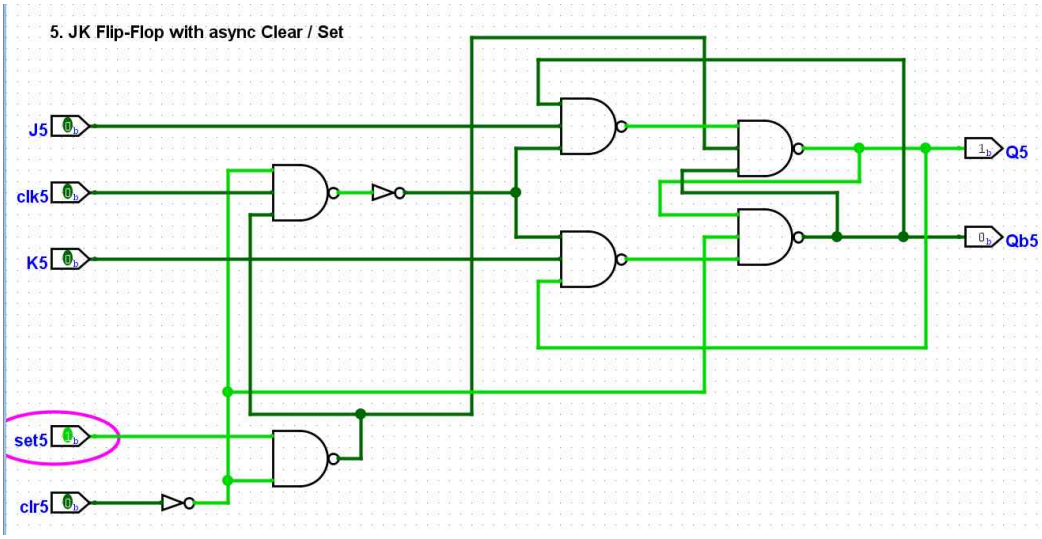


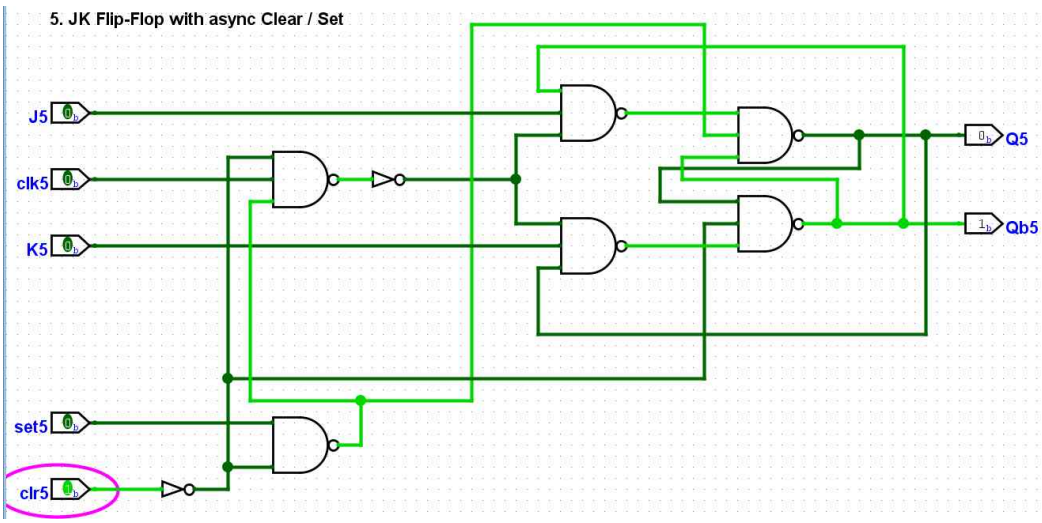



5. JK Flip-Flop with async Clear / Set
J5
clk5
K5
set5
clr5
Q5
Qb5

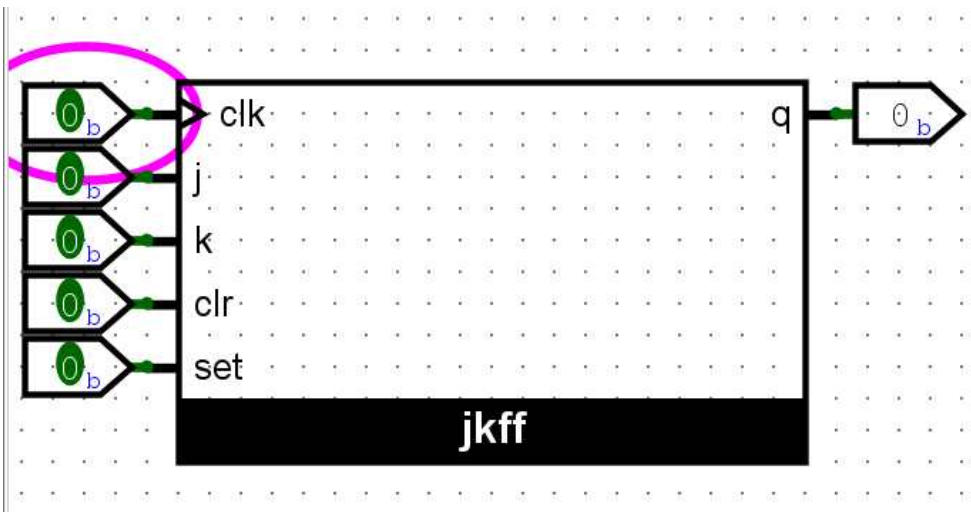

5. JK Flip-Flop with async Clear / Set
J5
clk5
K5
set5
clr5
Q5
Qb5

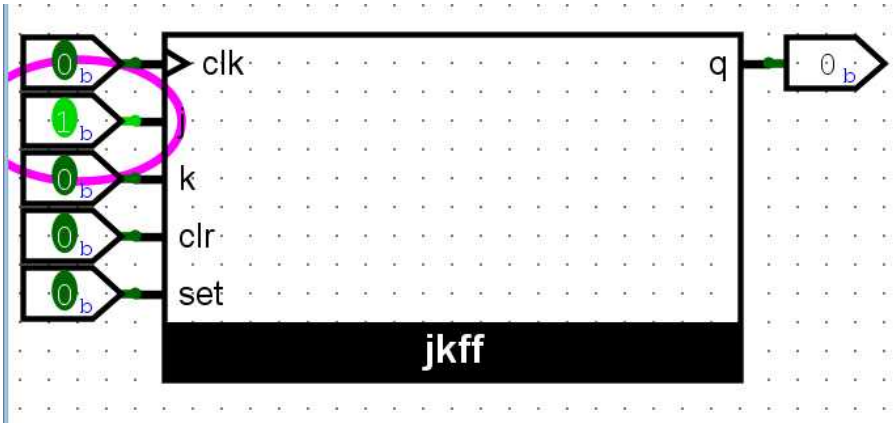

5. JK Flip-Flop with async Clear / Set
J5
clk5
K5
set5
clr5
Q5
Qb5

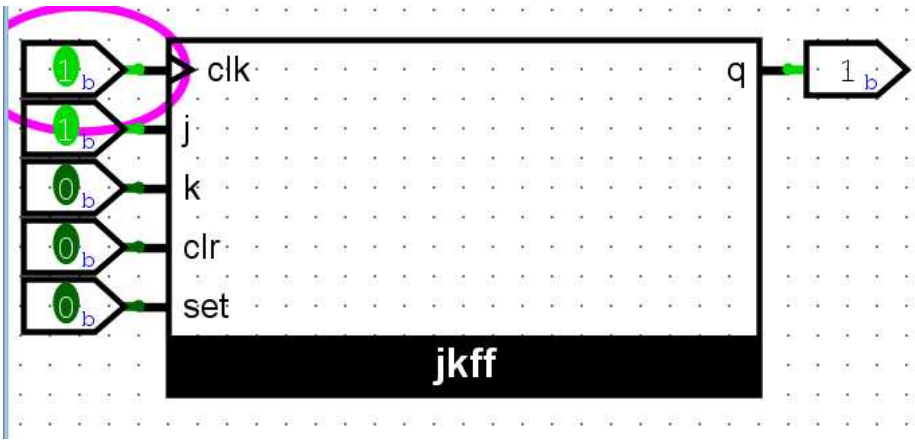

5. JK Flip-Flop with async Clear / Set
J5
clk5
K5
set5
clr5
Q5
Qb5

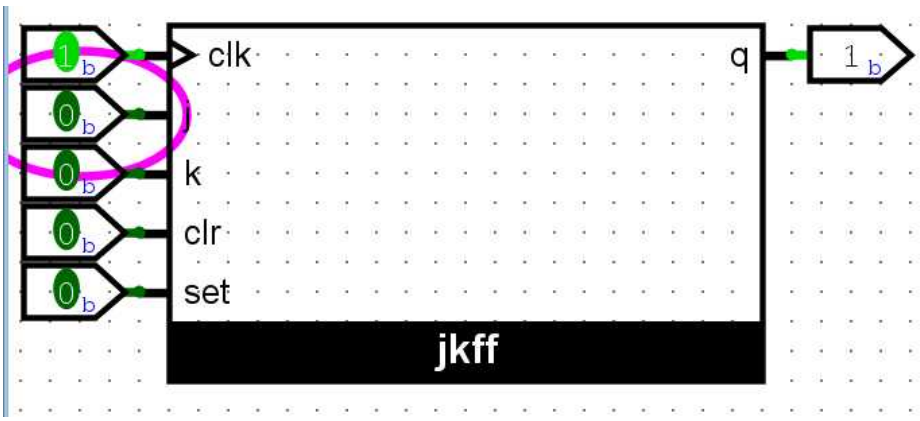

5. JK Flip-Flop with async Clear / Set
J5
clk5
K5
set5
clr5
Q5
Qb5

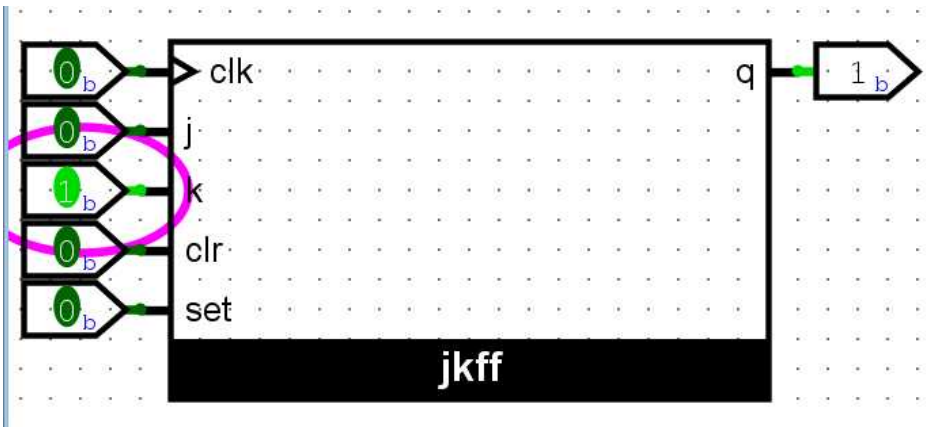
(2) VHDL-Based

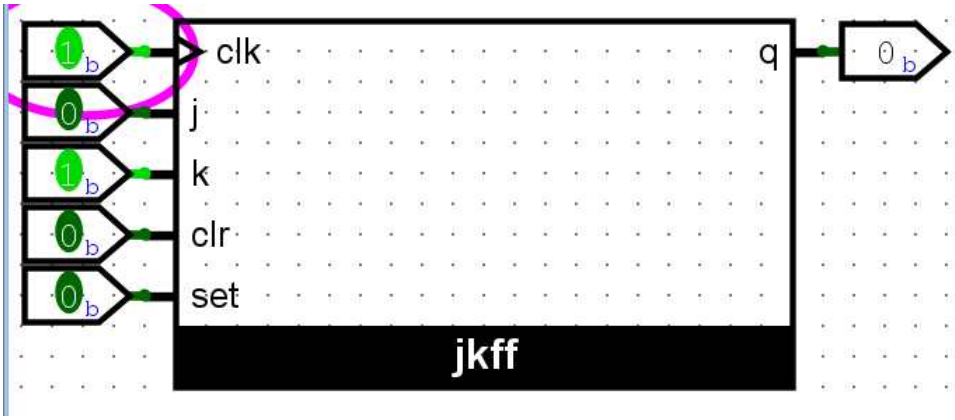

clk
j
k
clr
set
q
jkff

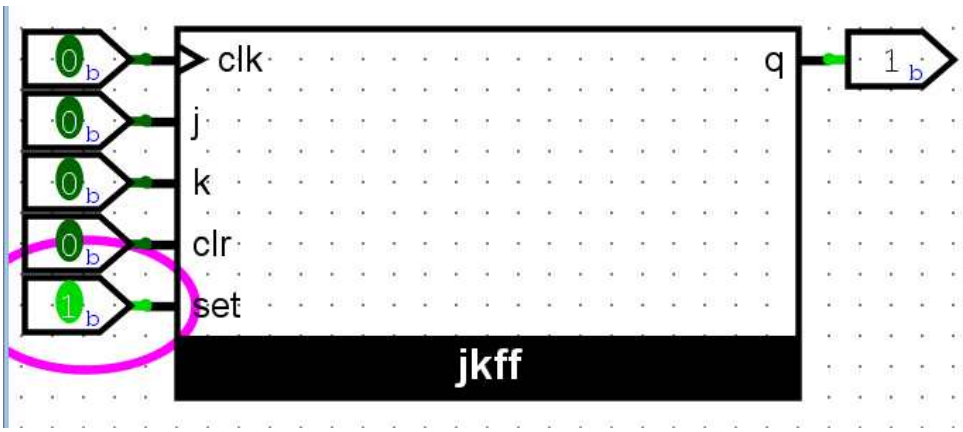

0 b
1 b
0 b
0 b
0 b
clk
k
clr
set
jkff
q
0 b

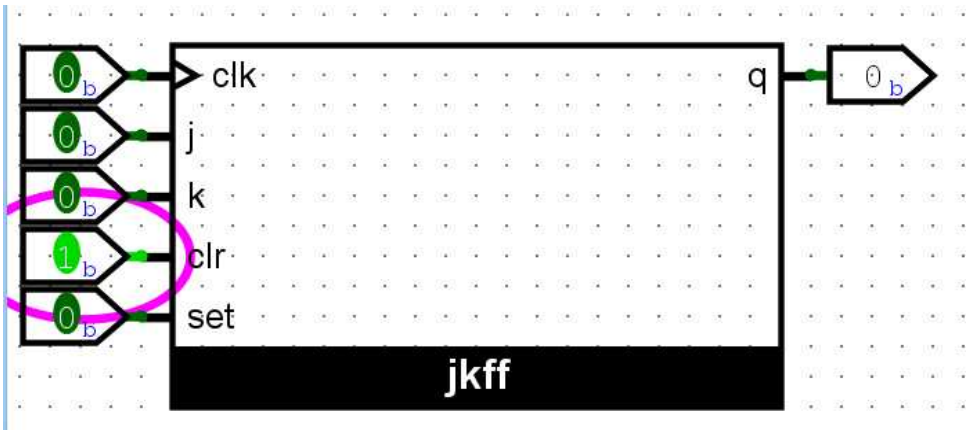

1 b
1 b
0 b
0 b
0 b
clk
j
k
clr
set
jkff
q
1 b

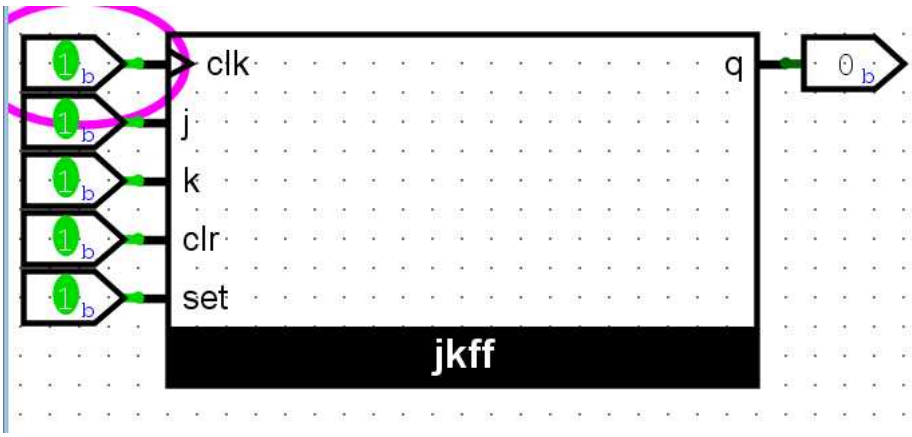

1 b
0 b
0 b
0 b
0 b
clk
j
k
clr
set
jkff
q
1 b


0 b
0 b
1 b
0 b
0 b
clk
j
k
clr
set
jkff
q
1 b


1 b
0 b
1 b
0 b
0 b
clk
j
k
clr
set
q
0 b
jkff


0 b
0 b
0 b
0 b
1 b
clk
j
k
clr
set
q
1 b
jkff


0 b
0 b
0 b
1 b
0 b
clk
j
k
clr
set
q
0 b
jkff


1 b
1 b
1 b
1 b
1 b
clk
j
k
clr
set
q
0 b
jkff

4. Result explanation

The simulation results show that all five circuits work as expected in both Logisim and VHDL. The latches and flip-flops keep their previous output through feedback. For the JK flip-flop, the output stayed in the error (E) state in Logisim at first, because Q and Q' are fed back into the input gates and there was no initial value. So a clear input was added to set Q to 0 at the start, and after that the circuit worked correctly.